

5_Lists in Python methods and manipulation

Lists are mutable and collection of different data elements and defined within square brackets.

```
mylist=[1,2,3,4,5]
print(mylist)
```

[1] ✓ 0.0s

... [1, 2, 3, 4, 5]

```
mylist2=[1,'a','3.5','Hanu']
print(mylist2)
```

[2] ✓ 0.0s

... [1, 'a', '3.5', 'Hanu']

```
print(mylist2[0])
```

[3] ✓ 0.0s

... 1

```
print(mylist2)
```

[4] ✓ 0.0s

... [1, 'a', '3.5', 'Hanu']



```
mylist2[0]=5 #List is mutable, we can change the value of an element in a list
```

[7] ✓ 0.0s

```
print(mylist2)
```

[6] ✓ 0.0s

... [5, 'a', '3.5', 'Hanu']

```
print(mylist2)
```

[6] ✓ 0.0s

... [5, 'a', '3.5', 'Hanu']

```
#List slicing
```

```
print(mylist2[0:3]) #Prints elements from index 0 to 2
```

[8] ✓ 0.0s

... [5, 'a', '3.5']

```
#List reversal
```

```
mylist2.reverse()
```

```
print(mylist2)
```

[9] ✓ 0.0s

... ['Hanu', '3.5', 'a', 5]

```
print(mylist2)
```

```
rev_list=mylist2[::-1] #Another way to reverse a list
```

```
print(rev_list)
```

[10] ✓ 0.0s

... ['Hanu', '3.5', 'a', 5]

[5, 'a', '3.5', 'Hanu']

```
# Modifying a list
```

```
print(mylist)
```

```
mylist.append(6)
```

```
mylist.remove(3)
```

```
print(mylist)
```

[13] ✓ 0.0s

... [1, 2, 3, 4, 5]

[1, 2, 4, 5, 6]

```
#concatenation of lists
```

```
mylist3 = mylist + mylist2
```

```
print(mylist3)
```

[14] ✓ 0.0s

... [1, 2, 4, 5, 6, 'Hanu', '3.5', 'a', 5]

```
#list repetition
mylist4 = [1] * 10
print(mylist4)
```

[16] ✓ 0.0s

... [1, 1, 1, 1, 1, 1, 1, 1, 1, 1]

```
print(len(mylist4)) #length of the list
```

[17] ✓ 0.0s

... 10

```
#List within a list or nested list
mylist5 = [[1, 2], [3, 4,5], [6,7,8,9], ['Hanu',37,'200k']]
print(mylist5)
```

[21] ✓ 0.0s

... [[1, 2], [3, 4, 5], [6, 7, 8, 9], ['Hanu', 37, '200k']]

```
#fetching an element from a list within a list
print(mylist5[1][2]) #fetching 3rd element from 2nd list
print(mylist5[3][0]) #fetching 1st element from 4th list
```

[24] ✓ 0.0s

... 5
Hanu

```
# List element manipulation
print(mylist2)
mylist2.append(7) #Appends 7 to the end of the list
mylist2.insert(2,10) #Inserts 10 at index 2
mylist2.pop(1) #Removes the element at index 1
mylist2.remove('a') #Removes the first occurrence of '3.5'
print(mylist2)
```

[30] ✓ 0.0s

... ['Hanu', 10, 10, 'a', 5, 7, 7, 7, 7]
['Hanu', 10, 10, 5, 7, 7, 7, 7, 7]

```
# List comprehension
squares = [x**2 for x in range(10)]
print(squares)
```

[31] ✓ 0.0s

... [0, 1, 4, 9, 16, 25, 36, 49, 64, 81]

